

HUJI Research Monitor

Weekly Digest · Week 30 · July 20, 2026

EXECUTIVE SUMMARY

This week's digest highlights a diverse portfolio of breakthroughs poised for significant commercial impact. Innovations span precision medicine, including novel diagnostics for breast cancer and AI tools for drug discovery, alongside transformative cell and gene therapies. We also feature advances in medical devices, such as robotic surgery and infection control coatings, and sustainable solutions for agriculture, underscoring Hebrew University's commitment to impactful research translation across critical global sectors.

#1

Pro-repair properties of a human embryonic stem cell-derived astrocyte cell therapy in demyelinating disorders.

36/50

Main Researcher: Tamir Ben-Hur

[Neuroscience] [Immunology] [Clinical] [Synthetic Biology]

"Off-the-shelf cell therapy to repair nerve damage offers new hope for demyelinating disorders like MS."

WHY NOW

Demyelinating diseases lack effective remyelination therapies. This scalable, clinical-grade cell therapy addresses a critical unmet medical need with high market potential.

COMMERCIAL ANGLE

Development of a scalable, off-the-shelf cell therapy for remyelination in Multiple Sclerosis and other demyelinating neurological disorders.

<https://pubmed.ncbi.nlm.nih.gov/42276062/>

#2

CBFB mutations predict endocrine therapy benefit in estrogen receptor-positive breast cancer.

35/50

Main Researcher: Albert Grinshpun

[Diagnostics] [Genomics] [Clinical]

"New biomarker predicts endocrine therapy response in ER+ breast cancer, personalizing treatment for patients."

WHY NOW

Companion diagnostics reduce ineffective treatments, saving costs and improving outcomes. This offers a clear path to market in precision oncology.

COMMERCIAL ANGLE

Development of a companion diagnostic test to personalize endocrine therapy selection for ER+ breast cancer patients, reducing ineffective treatments and improving clinical outcomes.

<https://pubmed.ncbi.nlm.nih.gov/42238456/>

#3

35/50

sc-rDSeq: a robust and cost-effective full-length total RNA sequencing method for single cells reveals multilayered heterogeneity in drug-resistant lung cancer cells.

Main Researcher: Oren Ram

[Genomics] [Diagnostics] [Drug Discovery] [Clinical]

"Cost-effective single-cell sequencing platform unveils hidden disease drivers for advanced diagnostics and drug discovery."

WHY NOW

Next-generation sequencing is a rapidly growing market. This platform's superior data breadth and cost-efficiency address critical needs in precision medicine.

COMMERCIAL ANGLE

The technology presents a high-value platform for next-generation precision diagnostics and drug discovery tools by providing more comprehensive transcriptomic data for identifying rare disease drivers.

<https://pubmed.ncbi.nlm.nih.gov/41978262/>

#4

34/50

Allele-specific antisense oligonucleotide treatment rescues

Main Researcher: Tamar Harel

[Drug Discovery] [Genomics] [Clinical]

"Precision ASO therapy selectively targets single-nucleotide mutations, offering safer treatments for genetic diseases."

WHY NOW

Antisense oligonucleotide (ASO) drugs are a validated therapeutic modality, and allele-specific targeting offers a significant leap in safety and efficacy for rare genetic disorders.

COMMERCIAL ANGLE

The ability to precision-target single-nucleotide mutations opens a high-value therapeutic pathway for rare genetic disorders and personalized genomic medicine.

<https://pubmed.ncbi.nlm.nih.gov/42239176/>

#5

33/50

The effect of a proprietary chlorine-stabilizing surface-coating formulation on healthcare environment disinfection in multidrug-resistant bacteria isolation rooms.

Main Researcher: Oren Zimhony

[Medical Device] [Materials] [Clinical]

"Durable antimicrobial coating dramatically reduces hospital superbug transmission, enhancing infection control protocols."

WHY NOW

Healthcare-associated infections and multidrug-resistant bacteria pose a persistent global threat, creating high demand for effective, scalable infection control solutions.

COMMERCIAL ANGLE

This technology offers a scalable solution for hospitals and long-term care facilities to enhance infection control protocols and reduce the transmission of multidrug-resistant organisms through high-durability antimicrobial coatings.

<https://pubmed.ncbi.nlm.nih.gov/42374536/>

#6

Multiple Charged Purine (MCP) PNA as a Simple Mode for Cellular Uptake.

33/50

Main Researcher: Eylon Yavin

[Drug Discovery] [Genomics] [Synthetic Biology]

"Breakthrough DNA analogue enters cells naturally, simplifying gene and antisense therapeutic delivery."

WHY NOW

Intracellular delivery remains a major bottleneck for genetic therapies. This innovative platform eliminates the need for expensive, complex delivery systems, improving drug development.

COMMERCIAL ANGLE

Development of a versatile, high-stability platform for intracellular genetic targeting and antisense therapeutics without the need for expensive lipid nanoparticles or viral vectors.

<https://pubmed.ncbi.nlm.nih.gov/42294022/>

#7

ASO Author Reflections: Robotic Cytorreduction at the Vascular Interface: Redefining Surgical Limits.

32/50

Main Researcher: Liron Kogan

[Medical Device] [Clinical] [Imaging]

"Robotic surgical system for precise tumor removal near blood vessels improves safety and expands treatability."

WHY NOW

The global robotic surgery market is rapidly expanding. This specialized platform addresses a challenging niche, reducing human error in complex oncological procedures.

COMMERCIAL ANGLE

Development of a high-precision robotic surgical platform for oncological cytorreduction to reduce human error and expand the treatability of complex vascular-adjacent tumors.

<https://europepmc.org/article/MED/42120689>

#8

Universal rhythmic architecture uncovers two modes of neural dynamics.

28/50

Main Researcher: Ayelet N Landau

[Diagnostics] [Medical Device] [Software/AI] [Neuroscience]

"Universal brain rhythm architecture enables personalized neurodiagnostics and advanced brain-computer interface development."

WHY NOW

Understanding brain activity is critical for diagnostics and BCI development. This foundational discovery enables more accurate and personalized neuro-technologies.

COMMERCIAL ANGLE

This universal architecture can be used to develop highly personalized neurodiagnostic software and advanced brain-computer interfaces that distinguish between stable cognitive states and rapid sensory responses.

<https://pubmed.ncbi.nlm.nih.gov/42218120/>

#9

RadD from *Fusobacterium nucleatum* engages NKp46 to promote antitumor cytotoxicity.

28/50

Main Researcher: Ofer Mandelboim

[Immunology] [Drug Discovery] [Clinical]

"Bacterial protein activates NK cells to fight tumors, paving the way for novel cancer immunotherapies."

WHY NOW

Immunotherapy is transforming cancer treatment. A novel mechanism for activating natural killer cells offers a promising new avenue for oncology drug development.

COMMERCIAL ANGLE

Development of RadD-based NK cell agonists or biomimetic immunotherapies to enhance natural killer cell cytotoxicity in oncology.

<https://europepmc.org/article/MED/42065370>

#10

Paenibacillus dendritiformis C454 as a sustainable biocontrol agent for management of bacterial plant diseases.

25/50

Main Researcher: Yael Helman

[AgriTech] [Synthetic Biology]

"Sustainable biological spray protects high-value crops, offering an eco-friendly alternative to chemical pesticides."

WHY NOW

There is a growing global demand for sustainable agriculture and organic produce, driving the market for effective biological pest control agents.

COMMERCIAL ANGLE

Development of a sustainable, biological foliar spray to replace chemical pesticides in large-scale vegetable and fruit production.

<https://europepmc.org/article/MED/41975582>

#11

Nitric oxide inhibition ameliorates cortical proteomic changes in the *Cntnap2*^{-/-} and *Shank3*^{Δ4-22} mouse models of autism spectrum disorder.

25/50

Main Researcher: Haitham Amal

[Drug Discovery] [Neuroscience] [Proteomics] [Genomics]

"Novel drug target for genetically-linked autism offers path to first-in-class small molecule treatments."

WHY NOW

Autism spectrum disorder represents a significant unmet medical need, and a validated molecular target for specific genetic subtypes holds immense therapeutic potential.

COMMERCIAL ANGLE

The identification of specific nitric oxide signaling pathways provides a validated pharmacological target for developing first-in-class small molecule treatments for genetically-linked autism spectrum disorders.

<https://europepmc.org/article/MED/42010670>

#12

24/50

Metabolic Endotoxemia Amplifies Estrogen Signaling Through Stromal Crosstalk in Obesity-Associated Breast Cancer.

Main Researcher: Michael Elkin

[Drug Discovery] [Immunology] [Clinical] [Diagnostics]

"Targeting inflammation in obese patients could treat hormone-responsive breast cancer more effectively."

WHY NOW

Obesity is a major risk factor for breast cancer, and this discovery offers novel therapeutic and diagnostic strategies for a growing patient population.

COMMERCIAL ANGLE

Development of targeted therapies or diagnostics that disrupt the endotoxin-induced stromal crosstalk to treat obesity-driven, hormone-responsive breast cancers.

<https://europepmc.org/article/MED/42196319>